Curriculum vitae

Karthikeyan Govindan

Associate Research Officer
The Wellcome Trust Research Laboratory,
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Experience

1. **Associate Research Officer -** Wellcome Trust Research Laboratory, Christian Medical College Vellore | Vellore, Tamil Nadu, India | 06/2026 – Present

2. **Research Officer -** Wellcome Trust Research Laboratory, Christian Medical College Vellore | Vellore, Tamil Nadu, India | 05/2022 – 05/2026

3. **Project Assistant** - King Institute of Preventive Medicine and Research | Chennai, Tamil Nadu, India 03/2021 – 02/2022

Research specialization

- Wastewater Genomic Surveillance
- Bacteriophage Surveillance
- Gut microbiome
- Pathogen sequencing
- Functional Genomics
- Bioinformatics

Publications

1. Gut microbiota in treatment-naïve active Crohn’s disease compared to those in remission: Divergent trends in mucosal versus fecal bacterial community. *Indian Journal of Gastroenterology.* 2026
2. Protease-Mediated Recovery of Amnion from Placenta and Its Application as a Bioprocessed Therapeutic Biomaterial for Wound Healing. *Journal of Health Science and Medical Research.* 2026
3. Exploring Gut Mucosal Microbiota to Distinguish Crohn's Disease from Intestinal Tuberculosis. *Inflammatory bowel diseases.* 2026
4. Dynamics of tissue and fecal microbiota in active Crohn’s disease and their ability to predict disease state. *World Journal of Gastrointestinal Pathophysiology.* 2025
5. AmpliconTyper-a tool for analysing ONT multiplex PCR data from environmental and other complex samples. *Microbial Genomics.* 2025
6. EnviroAmpDesigner–a tool for designing high specificity multiplex PCR primer panels for detecting and subtyping a target organism in environmental surveillance samples. b*ioRxiv* (2025).
7. Unveiling the Global Surge of Mpox (Monkeypox): A comprehensive review of current evidence. The Microbe. 2024.
8. qPCR based multipathogen detection for SARS-CoV-2, CrAssphage and Hepatitis E virus from wastewater samples. Protocols.io. 2024
9. Distribution and Functional Analyses of Mutations in Spike Protein and Phylogenetic Diversity of SARS-CoV-2 variants emerged during 2021 in India. Journal of Global Infectious Diseases. 2023.
10. Epitope identification and design of a potent multi-epitope vaccine construct against SARS-CoV-2 including emerging variants. Journal of Global Infectious Diseases. 2022.
11. Serum levels of matrix metalloproteinases as prognostic markers for severe dengue with plasma leakage. Experimental and Molecular Pathology. 2022.
12. The utility of inflammatory and endothelial factors in the prognosis of severe dengue. Immunobiology. 2022.
13. Phylodynamic pattern of genetic clusters, paradigm shift on spatio-temporal distribution of clades, and impact of spike glycoprotein mutations of SARS-CoV-2 isolates from India. Journal of Global Infectious Diseases. 2021.

Awards & Scholarships

1. **Bursary Awardee,** *Wellcome Connecting Science — Understanding Life: Using Large-Scale Biodiversity Reference Genomes* (2025)
2. **Bursary Awardee**, *Oxford Nanopore Technologies* (2025)

Current Research Projects

1. Novel tools for measuring typhoid vaccine impact (R01) [Funding - NIH].
2. Optimisation of an integrated protocol for the molecular detection and sequencing of priority viral and bacterial pathogens in environmental samples [Funding - BMGF]
3. Standardization of methods and models for multipathogen detection from wastewater in Vellore [Funding - BMGF]
4. Elimination of Malnutrition through Action-Research on Moderately Acute Malnourished Children (CSR Grant)A - Sun Pharma Ltd.
5. Vellore Typhoid ES for TCV impact assessment [Funding - BMGF]
6. Establishing sewage surveillance and laboratory testing methods for the environmental surveillance of COVID-19 [Funding - BMGF]
7. Long term stability of gut microbiota in patients with Crohn’s disease and correlation with biomarkers of inflammation and gut permeability [Funding - DST India]
8. Genetic attributes and virulence of Salmonella Typhi impacting its intracellular survival within Acanthamoeba spp. - Grand Challenges Explorations grant [Funding - BMGF]
9. Estimation of Rotavirus Evolutionary Divergence of Time Using Bayesian Evolutionary Analysis (BEAST).

Completed Research Projects

1. Evaluation and characterization of crucial biomarker profiles for stratifying Dengue disease severity [Funding - ICMR]
2. Diagnosis of SARS-CoV-2 and detection of other viral respiratory pathogens in SARI patients.
3. Spatiotemporal analysis and deep mutational scanning of Indian SARS-CoV-2 genomes.
4. Establishment of human primary Gingival fibroblast and primary Dermal fibroblast cell lines.

Education

King Institute Of Preventive Medicine And Research | Chennai, Tamil Nadu, India

Master of Science (MSc) - Molecular Virology | 07/2021 | Percentage: 75.5

Vels Institute Of Science Technology And Advanced Science | Chennai, Tamil Nadu, India

Bachelor of Science (BSc) - Microbiology | 07/2018 | CGPA: 7.88

Presentations

1. Chromosome scale reference genome model - Acanthamoeba polyphaga (ATCC-30461) – T4 genotype, *Understanding Life: Using Large-Scale Biodiversity Reference Genomes, Wellcome Connecting Science (October 2025)*

2. Functional Characterization and Reference Genome Assembly of *Acanthamoeba polyphaga* (ATCC-30461) – T4 genotype, BioDiversity Genomics Conference 2024 (BG24), Tree of Life Programme, Wellcome Sanger institute. October 2024.

3. Composition and Temporal Dynamics of Microbial Communities in Drinking Water from Peri-Urban Areas of Vellore. Winter Symposium, Christian Medical College and Hospital, Vellore. (February 2024).

Skills

Technical skills

- Sequencing - MiSeq & NextSeq1000 (Illumina),
- MinION (ONT)
- PCR- Conventional PCR , Real time PCR.
- ELISA.
- Nucleic Acid extraction.
- Gel electrophoresis.
- Qubit Fluorometer/spectrophotometry.

- Tapestation (Fragment analyzer).
- Fluorescence Microscope.
- Cell Culture / Virus Isolation

Bioinformatics Skills

- NGS QC: FastQC, MultiQC, FastP, Trimmomatic, Quast.
- Genome Assembly: SPAdes, Flye.
- Genome Binning: MetaBAT2, MetaWrap, MetaCoAg, Maxbin.
- Genome Annotation: Prokka, Augustus, BRAKER3, Artemis.
- Functional Annotation: KEGG, KO, GO, eggNOG, InterPros, AlphaFold.
- Transcriptomics: Hisat2, StringTie, DESeq2, Cufflinks.
- Phylogenetics: RAxML, FastTree, BEAST, MEGA.
- Metagenomics: Qiime2, DADA2, metaSPAdes. Megahit.
- Classifier: 16s and 18s Silva and PR2 classifiers.
- Workflow management: Nextflow, Snakemake and Bash.
- Dashboards & Real-time Visualization**:** Shiny | GitHub Pages | ARTIC RAMPART

Computing Skills

- High Performance Computing(HPC) - Google Compute Engine | AWS cloud computing
- Linux – CPU | GPU parallelization
- Coding language - Bash | R
- Google Colab Notebook (Python and R/bash)
- GitHub - version control Codes | Documents | Data
- Containerization - Docker | Pipelines | Scripting | Automation
- Document management – R Markdown | Quarto | GitBook
- Web servers: Chan Zuckerberg ID | Galaxy servers | BV-BRC

- Web site development : Github pages.

Certificates

- Certificate of Appreciation for Organizing the *Galaxy Training Academy 2025*.
- National Workshop on Metabolic Modelling: From Pathways to Predictions 2026 (IIT Madras)
- Analyzing and Visualizing NextGen Reads with Magic-BLAST (NIH- NCBI)
- Bioinformatics for Biologists - An Introduction to Linux, Bash Scripting and R (Wellcome connecting science).
- Eukaryotic Genome Assembly - How to use BlobToolKit for quality assessment (Wellcome connecting science).
- EMBL-EBI- Training online Courses.
- A Practical guide for SARS CoV-2 Whole Genome Sequencing (Wellcome connecting science).
- From swab to server: Testing , Sequencing and Sharing During a Pandemic (Wellcome connecting science).
- Bacterial Whole Genome Sequencing and Advanced Genomic Data Mining.
- Nanopore Community Meeting (NCM) 2022.
- Second virtual training on KFD virus Biosafety, molecular and serological methods (NIV-pune).
- Trained on vector virus relationship, Epizootiology of vector borne viral diseases, Molecular Entomology, Mosquito species identification, collection, processing , preservation and dissection of mosquito. (ICMR-VCRC).

Contact info

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6. **Google Scholar** ID - https://scholar.google.com/citations? hl=en&user=is4cNvYAAAAJ